

7th Grade Unit 5 Assessment Guide

General Information

Unit 5 uses the previously learned concepts of probability, simulation, area, and perimeter to build students' understanding of proportional reasoning. One of the areas of emphasis of Grade 7 is proportional reasoning. Units 6 and 7 have a stronger focus on proportional reasoning, but Unit 5 allows students to begin applying proportions in simpler contexts related to their prior learning. Unit 5 is an example of where the content spirals through previously learned content and builds on students' prior knowledge.

Teachers should look for opportunities to reinforce the benchmark MA.7.DP.2.4, which serves as a supporting benchmark throughout the first four lessons of the unit. Teachers can use question 5 to support students who show a misunderstanding of experimental and/or theoretical probability. Teachers may find that students who struggle with proportional reasoning need to revisit determining probability.

The proportional reasoning used in MA.7.DP.1.3 is similar to the proportional reasoning used in MA.7.GR.1.5. Students may benefit from discussing the similarities of using the proportional relationship of a random sample to the scale factor in problems that involve dimensions and areas of geometric figures.

Teachers should monitor how students answer question 7. Students who need to find the perimeter and/or area of both figures have not yet understood that the perimeter is one-dimensional just as a side length is one-dimensional, so the scale factors are the same. Students who miss question 7b may need further support in building an understanding of area as two-dimensional. Teachers may want to build that understanding using grid paper.

At the beginning of the assessment is a box with the statements:

Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit, show the operation that you entered into the calculator and the result. It is helpful if you label the operation with information from the word problem.

This suggestion can also help a student self-scaffold a problem. As you consider the point distribution, consider the sub-steps students may take. When you set up the point distribution consider that not every student needs to show sub-steps so your point distribution should not punish students who can think through the problem and find an answer using a calculator such as the Desmos scientific calculator. It is possible that students can enter these problems as one expression into a calculator such as the Desmos scientific calculator to get the correct answer.

Calculator Information	The questions on this assessment are calculator.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1	Benchmark(s)	MA.7.GR.1.5
640		Recommended Scoring (8 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	2a	Benchmark(s)	MA.7.DP.1.3
Middle schoolers at Sebastian Middle School and cell phone owners		Recommended Scoring (6 Total Points)	Consider giving students 3 points for each correctly named population.
Question	2b	Benchmark(s)	MA.7.DP.1.3
○ Both surveys ○ Neither survey ○ The survey conducted by a cell phone company ○ The survey conducted at Sebastian Middle School would be a representative sample of all cell phone users.		Recommended Scoring (6 Total Points)	No partial credit is suggested.

Question	3	Benchmark(s)	MA.7.DP.1.3
1250		Recommended Scoring (8 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	4a	Benchmark(s)	MA.7.DP.1.3
2918		Recommended Scoring (8 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	4b	Benchmark(s)	MA.7.DP.1.3
11904		Recommended Scoring (8 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	5a	Benchmark(s)	MA.7.DP.2.4
$\frac{15}{90}$ or $\frac{3}{18}$ or $\frac{1}{6}$		Recommended Scoring (8 Total Points)	Consider partial credit for a correct probability that is given in a format other than a fraction.

Question	5b	Benchmark(s)	MA.7.DP.2.4
$\frac{9}{90}$ or $\frac{3}{30}$ or $\frac{1}{10}$		Recommended Scoring (8 Total Points)	Consider partial credit for a correct probability that is given in a format other than a fraction.
Question	5c	Benchmark(s)	MA.7.DP.2.4
78		Recommended Scoring (8 Total Points)	No partial credit is suggested.
Question	5d	Benchmark(s)	MA.7.DP.2.4
128		Recommended Scoring (8 Total Points)	No partial credit is suggested.
Question	6	Benchmark(s)	MA.7.GR.1.5
200		Recommended Scoring (8 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	7a	Benchmark(s)	MA.7.GR.1.5
3.5 or $\frac{7}{2}$		Recommended Scoring (8 Total Points)	No partial credit is suggested.

Question	7b	Benchmark(s)	MA.7.GR.1.5
3.5^2 or 12.25 or $\left(\frac{7}{2}\right)^2$ or $\frac{49}{4}$		Recommended Scoring (8 Total Points)	Consider partial credit for a student who may have the wrong scale factor in part a but gives a correct scale factor for area based on their incorrect answer.